

Cancer incidence in South Asian migrants to England, 1986–2004: Unraveling ethnic from socioeconomic differentials

Camille Maringe¹, Punam Mangtani², Bernard Rachet¹, David A. Leon³, Michel P. Coleman¹ and Isabel dos Santos Silva³

¹ Cancer Research UK Cancer Survival Group, Department of Non-Communicable Disease Epidemiology, Faculty of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, London, United Kingdom

² Faculty of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, London, United Kingdom

³ Department of Non-Communicable Disease Epidemiology, Faculty of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, London, United Kingdom

Studies on cancer in migrants are informative about the relative influence of environmental and genetic factors on cancer risk. This study investigates trends in incidence from colorectal, lung, breast and prostate cancer in England among South Asians and examines the influence of deprivation, a key environmental exposure. South Asian ethnicity was assigned to patients recorded in the population-based National Cancer Registry of England during 1986–2004, using the computerized algorithm SANGRA: South Asian Names and Groups Recognition Algorithm. Population denominators were derived from population censuses. Multivariable flexible (splines) Poisson models were used to estimate trends and socioeconomic differentials in incidence in South Asians compared to non-South Asians. Overall, age-adjusted cancer incidence in South Asians was half that in non-South Asians but rose over time. Cancer-specific incidence trends and patterns by age and deprivation differed widely between the two ethnic groups. In contrast to non-South Asians, lung cancer incidence in South Asians did not fall. Colorectal and breast cancer incidence rose in both groups, more steeply in South Asians though remaining less common than in non-South Asians. The deprivation gaps in cancer-specific incidence were much less marked among South Asians, explaining some of the ethnic differences in overall incidence. Although still lower than in non-South Asians, cancer incidence is rising in South Asians, supporting the concept of transition in cancer incidence among South Asians living in England. Although these trends vary by cancer, they have important implications for both prevention and anticipating health-care demand.

Longstanding studies on cancer in migrants highlight the predominant influences of lifestyle rather than genetic factors on cancer risk.^{1–3} Cancer incidence is also strongly associated with socioeconomic group,² which varies between ethnic groups.⁴ In US studies, ethnicity is commonly used as a proxy for social disadvantage, but these variables are not equivalent in the United Kingdom.³ With a few exceptions,^{5–7} very little work in general has been done to

attempt to separate out the effects of deprivation from ethnicity.

South Asians in the United Kingdom are the largest group (4% of the population in 2001) to have moved from a cancer incidence setting with lower incidence than western populations.⁸ They also have different socioeconomic distributions.⁴ With the greater numbers of working age adults migrating to England around the 1960s,⁹ they have now reached ages at which cancer risk is appreciable. Recent mortality from specific cancers remains low in South Asians than in non-South Asians.¹⁰ Underlying cancer incidence patterns are however required to assess the relative importance of environmental factors and treatment effects. We also examine socioeconomic disparities in cancer between South Asians and non-South Asians at national level.

Key words: cancer incidence, South Asians, migrants, deprivation, trends

Additional Supporting Information may be found in the online version of this article.

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Correspondence to: Punam Mangtani, Faculty of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, Keppel Street, WC1E 7HT London, UK, Tel.: +44-207-7927-2057, Fax: +44-20-7436-4230, E-mail: punam.mangtani@lshtm.ac.uk

Material and Methods

Data sources

Cancer incidence. We used cancer incidence registrations in the National Cancer Registry for England at the Office for National Statistics (ONS) for 3,230,279 persons diagnosed with a malignant neoplasm between 1986 and 2004. We limited the study to adults aged 15–84 years at diagnosis because of the lack of reliable population estimates for older South Asians.

What's new?

Looking at cancer rates among immigrant populations – and how they change over time – can highlight the effects of lifestyle on cancer risk. This study followed South Asians in England over two decades to evaluate their cancer rates, and to separate genetic effects from quality of life influences. South Asians suffered less cancer than others, but the incidence is rising. The report also emphasizes that ethnicity and socio-economic deprivation each affect cancer risk independently, and that deprivation has less impact on cancer risk in South Asians than other ethnic groups.

South Asians have distinctive names, and validated computerized name algorithms have successfully identified them in individual-level data¹¹ over a much longer period than is possible with hospital data. Ethical and statutory approvals were obtained to flag the names on cancer registration records at the Office for National Statistics for South Asian ethnicity. The South Asian Names and Groups Recognition Algorithm (SANGRA)¹¹ was used to ascertain South Asian ethnicity. The algorithm incorporates directories of South Asian first names and surnames together with their religious and linguistic origin. Both visual inspection, as in similar studies¹² and exclusion of common Muslim names from countries outside of South Asia compiled from death certificate data¹⁰ were used to reduce further the number of false positives. A total of 31,187 South Asian adults (15–84 years) with cancer were identified representing 1% of all adult patients.

We analyzed separately the four most common malignancies: colorectum (ICD9: 153, 154; ICD10: C18–C21), lung (ICD9: 162; ICD10: C33, C34), breast (women, ICD9: 174; ICD10: C50) and prostate (ICD9: 185; ICD10: C61) and all malignant neoplasms combined (ICD9: 140–208, ICD10: C00–C97).

Population data. To estimate South Asian and non-South Asian population sizes by age and single year of diagnosis, age-sex population structures were obtained from the 1991 and 2001 UK Censuses. South Asians were defined according to the census categories “Indian,” “Pakistani” and “Bangladeshi,” together with “Other groups: Asian” in the 1991 census and “Other Asians” in the 2001 census (mixed White and Asian were not considered). Despite changes in the labeling of ethnic categories between the two censuses, these were shown to be comparable between the two censuses.¹³ Mid-year population estimates by ethnicity were provided by ONS only for the years 2002–2004. Between the years 1991 and 2001, yearly age-specific population estimates for South Asians were obtained by linear regression. The South Asian population size by age and sex in 1991 was assumed to be the same in the years 1986–1990 as South Asian population estimates from the UK Government's Labour Force survey were stable for that period.¹⁴ Non-South Asian population figures were obtained by subtracting the South Asian population estimates from the mid-year estimates as well as the 1991 and 2001 census figures available for the general population.

Deprivation data. To estimate South Asian and non-South Asian population sizes by age, deprivation and single year of diagnosis, we used deprivation measured using the Townsend

score,¹⁵ a small area-based measure of deprivation derived from the patient's residence at diagnosis. Importantly, for our purposes, no information on ethnic composition of areas is used in the construction of the score. Deprivation categories are the quintiles of the distribution of all the area-specific Townsend scores in England. We used the 1991-Townsend at Census Enumeration District level (mean population 500) over the period 1986–1995 and the 2001-Townsend at Lower Super Output Area level (mean population 1,500, but more socially homogeneous)¹⁶ for the period 1996–2004. The use of small and socially homogeneous area deprivation category is likely to reflect the socioeconomic status of our study population. Population by deprivation category were obtained from commissioned tables from Census data of the South Asian population by sex and 5-year age group for each Enumeration District (in 1991) or Lower Super Output Area (2001) from ONS.

Statistical methods

To reduce noise from small numbers, we smoothed the South Asian population estimates across 5-year age groups in 1991 and 2001–2004 using regression models with restricted cubic splines.¹⁷ Directly, age-standardized rates were calculated using the world standard population in 15–74-year olds.¹⁸

Multivariable Poisson regression with cubic splines¹⁷ was used to model cancer-specific incidence rates over time and by ethnicity. In a first model, where the outcome was incidence rates, the explanatory factors were year of diagnosis, age at diagnosis and ethnicity. Incidence rate ratios (IRRs) in South Asians relative to those in non-South Asians were used for each year to summarize temporal changes in cancer incidence.

In a second model, we included deprivation as an extra explanatory factor. To examine deprivation gradients, age patterns and temporal trends in incidence in South Asians compared to non-South Asians, we fitted interaction terms between deprivation and ethnicity, age and ethnicity and year of diagnosis and ethnicity, respectively. The importance of each factor was assessed using log-likelihood ratio tests. The incidence rates in 50-years-old were obtained from these models to illustrate trends graphically. We also calculated *p*-values for the homogeneity of the deprivation gradients between ethnic groups to assess the importance of the interaction between deprivation and ethnicity. Increasing incidence with increasing deprivation is reported as a positive deprivation gradient. A negative deprivation gradient is defined by decreasing incidence with increasing deprivation.

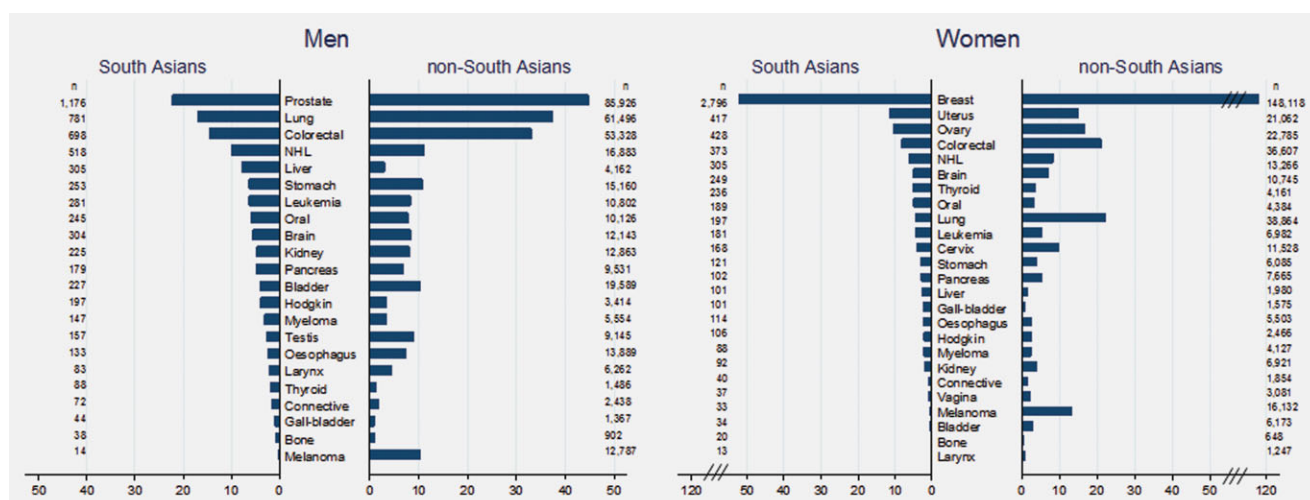


Figure 1. 1999–2004 age-standardized incidence rates up to 74 years, per 100,000 (and number of cases). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

p-Values for linear trend in incidence with deprivation by ethnic group were calculated from separate multivariable Poisson regression models in which deprivation was modeled as a continuous variable.

Results

In 2001, the South Asian population was on average 30-years-old, 10 years younger than non-South Asians. The mean age at cancer diagnosis was 60 and 55 years for South Asians men and women and 67 and 64 years for non-South Asians. Approximately, half the South Asian population and its cancer cases were in the most deprived quintile of the Townsend index (data not presented).

In patients aged 15–74 years, cancers of the prostate, lung and large bowel (colorectum) were the three most common cancers in both South Asian and non-South Asian men, but age-standardized incidence in South Asians was half that of non-South Asians (Fig. 1). Cancers of the breast, uterus and ovary were the most common among South Asian women, while cancers of the breast, lung and colorectum were the most common in non-South Asians (Fig. 1).

All-cancer incidence (for patients aged 15–84 years at diagnosis) rose steadily over time in both South Asian and non-South Asian men and women, but more rapidly among South Asians; thus, the ratio of incidence rates in South Asians over non-South Asians (IRR) moved closer to one (from 0.47 and 0.37–0.58 and 0.55 for men and women, respectively; Table 1 and Figs. 2 and 3). For all neoplasms combined and for all other cancer sites, except lung cancer in women, the incidence patterns with time were significantly different between South Asians and non-South Asians (p -value < 0.001, Table 1 and Supporting Information Table S1).

Colorectal cancer incidence rose more quickly in South Asians, but was still 40% lower than in non-South Asians in 2004 (Figs. 2 and 3, data in Supporting Information

Table S1). The trend with age was less marked in South Asians, while incidence rates in the youngest were comparable in both ethnic groups (data not presented).

The incidence rates for lung cancer among South Asian men remained stable between 1986 and 2004 (Fig. 2). A fall in incidence over time in their non-South Asian counterparts led to an increasing IRR for ethnicity over time (from 0.34 in 1986 to 0.49 in 2004, Fig. 3, Supporting Information Table S1). Lung cancer incidence increased with time at a similar pace for South Asian and non-South Asian women (Fig. 2) with no significant differences in the IRR over time (p value > 0.05, Supporting Information Table S1).

Incidence rose for breast and prostate cancers over the period 1986–2004 in both ethnic groups. Incidence remained substantially lower among South Asians throughout, but generally rose faster than among non-South Asians. As a result, the IRRs for ethnicity moved closer to one over time from 0.4 to 0.6 for breast and prostate cancers (Figs. 2 and 3 and Supporting Information Table S1). Age patterns were similar in both ethnic groups for prostate cancer. South Asian women showed a plateau in breast cancer incidence after age 45–50 (data not shown).

The deprivation gradient in all-cancers incidence was more marked in non-South Asian than South Asian men (Table 1 and Fig. 4). A gradient was present for non-South Asian women, but none was observed for their South Asian counterparts. For both ethnic groups, men showed stronger deprivation gradients than women, for all sites except lung cancer. There was no overlap between the two ethnic-specific distributions, with incidence in all deprivation categories in South Asians being far lower than in the most affluent non-South Asians (Fig. 4). Including deprivation in the model slightly reduced the gap in incidence between ethnic groups (Table 1 and Fig. 3).

The positive deprivation gradient in lung cancer incidence was less pronounced among South Asians, with no obvious pattern among South Asian women (Fig. 4), p -value test-for-trend = 0.95 (Supporting Information Table S1). Nevertheless, incidence in the most deprived South Asians remained below the level of the most affluent non-South Asians (Fig. 4). Adding deprivation to the models reduced the appa-

rent protective effect of South Asian ethnicity for lung cancer in women (Fig. 3).

The incidence of breast cancer (women) and prostate cancer was higher among affluent groups in both ethnic groups. This negative gradient was more obvious in South Asians for both cancers (all p -test-for-trend < 0.001; Fig. 4). When deprivation was taken into account, the IRRs for ethnicity moved closer to

Table 1. Cancer incidence rate ratios [IRR, with 95% confidence intervals (CI)]^{1,2} comparing year of diagnosis and deprivation categories separately in South Asians and non-South Asians aged 15–84 years,³ comparing South Asians to non-South Asians by sex: all neoplasms

	South Asians			Non-South Asians			South Asians/ non-South Asians			<i>p</i> -Value ⁴
	IRR ¹	95%	CI	IRR ²	95%	CI	IRR ³	95%	CI	
Men										
<i>All malignant neoplasms</i>										
Model including year and age at diagnosis										
1986	1				1		0.47	0.45	0.49	<0.001
1995	1.36	1.33	1.39	1.22	1.22	1.22	0.52	0.51	0.53	
2004	1.67	1.59	1.75	1.34	1.33	1.34	0.58	0.57	0.60	
Model including year, age at diagnosis and deprivation										
1986	1				1		0.50	0.48	0.51	
1995	1.49	1.48	1.50	1.27	1.27	1.27	0.58	0.57	0.60	
2004	1.70	1.69	1.71	1.37	1.37	1.37	0.62	0.60	0.63	
Least deprived	1			1						<0.001
2	1.01	0.98	1.04	1.07	1.07	1.07				
3	0.99	0.96	1.02	1.17	1.17	1.18				
4	1.02	0.99	1.04	1.28	1.27	1.28				
Most deprived	1.12	1.10	1.15	1.36	1.36	1.37				
<i>p</i> -Value for linear trend	<0.001			<0.001						
Women										
<i>All malignant neoplasms</i>										
Model including year and age at diagnosis										
1986	1				1		0.37	0.35	0.39	<0.001
1995	1.79	1.76	1.82		1.24	1.24	0.53	0.53	0.54	
2004	2.03	1.99	2.06		1.36	1.36	0.55	0.54	0.56	
Model including year, age at diagnosis and deprivation										
1986	1				1		0.43	0.42	0.45	1.00
1995	1.83	1.81	1.84		1.29	1.29	0.61	0.60	0.63	
2004	2.01	2.00	2.02		1.38	1.38	0.63	0.62	0.64	
Least deprived	1				1					<0.001
2	0.93	0.90	0.96		1.04	1.04	1.05			
3	0.95	0.93	0.98		1.11	1.11	1.11			
4	0.91	0.89	0.93		1.16	1.16	1.16			
Most deprived	0.93	0.91	0.95		1.18	1.18	1.18			
<i>p</i> -Value for linear trend	<0.001			<0.001						

¹Incidence rate ratios for the effect of year of diagnosis or of deprivation among South Asians. ²Incidence rate ratios for the effect of year of diagnosis or of deprivation among non-South Asians. ³Incidence rate ratios comparing South Asians to non-South Asians, by year of diagnosis or by deprivation. ⁴Test for heterogeneity in incidence trends by year of diagnosis or deprivation between ethnic groups.

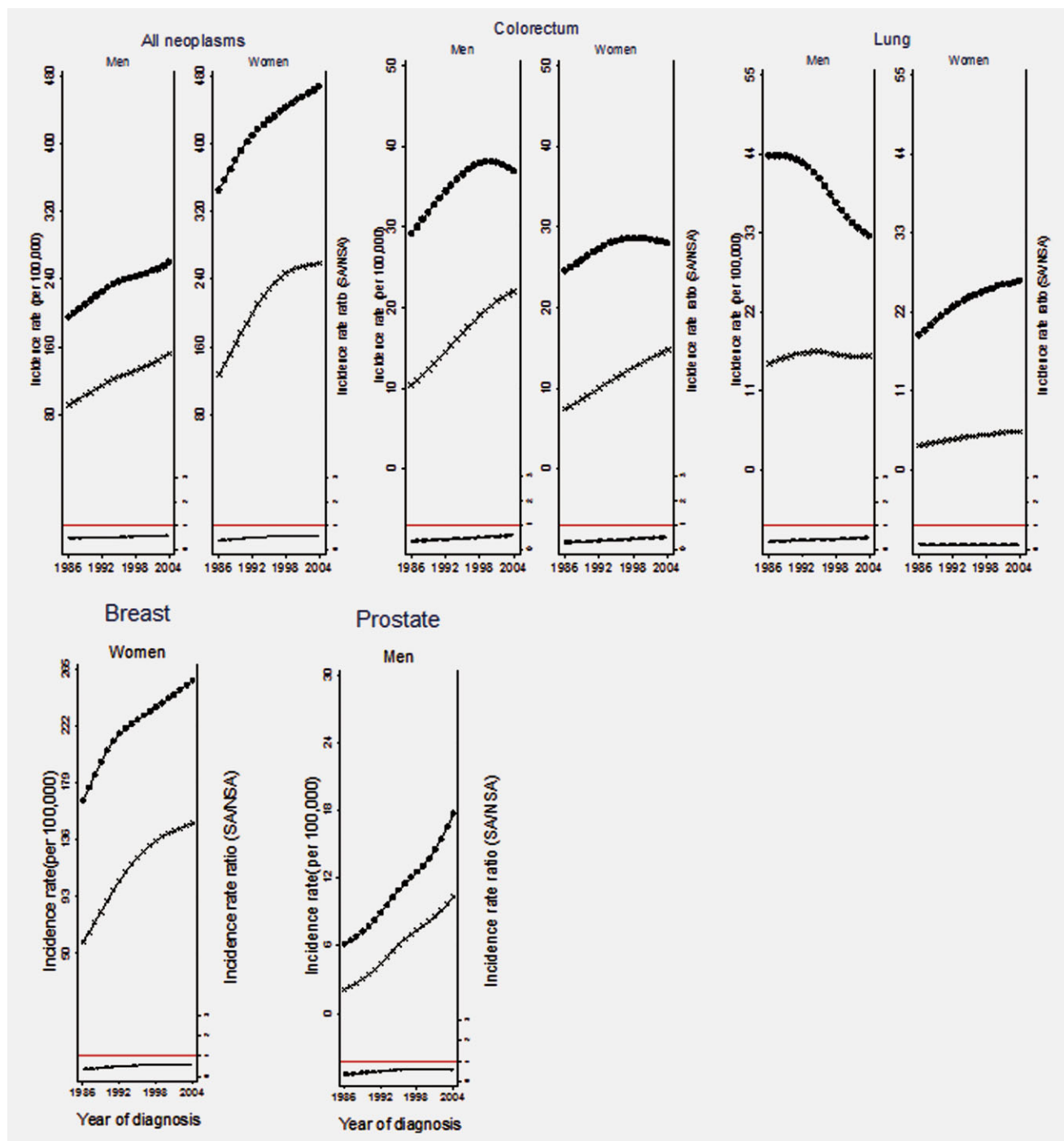


Figure 2. Modeled cancer incidence rates by ethnic group and incidence rate ratios in South Asians (X) compared to non-South Asians (circles), by cancer, year of diagnosis and sex in 50-year olds. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

one, that is, the protective effect of ethnicity was less than that estimated without accounting for deprivation (Fig. 3).

No deprivation gradient was seen for colorectal cancer incidence in South Asian women (p -test-for-trend: 0.36), and an inverse gradient was observed in South Asian men (p -test-for-trend < 0.001), a marked contrast to the positive gradient in non-South Asians (Fig. 4). For all cancers, except colo-

rectal cancer in women, the deprivation gradients are not homogeneous between ethnic groups (p -test-for-heterogeneity < 0.001), and incidence in the most deprived South Asian categories was up to 50% lower than in the most affluent non-South Asians (Fig. 4). Addition of deprivation in the models reduced the apparent beneficial effect of South Asian ethnicity, that is, part of the effect of "ethnicity" reflected the

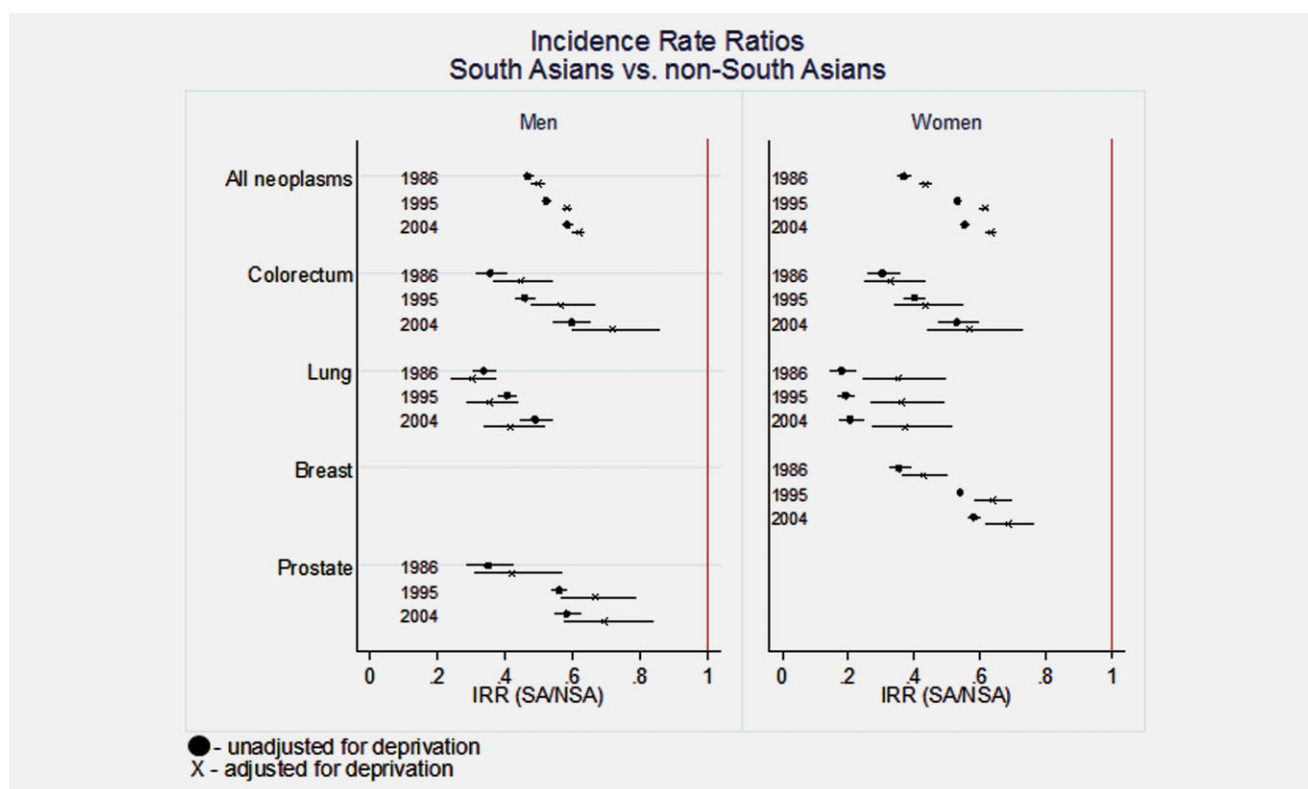


Figure 3. Impact of adjusting for deprivation on the trends in cancer incidence rate ratios by cancer site: South Asians *versus* non-South Asians aged 15–84 years. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

fact they are, on average, more deprived than non-South Asians with a lower risk of colorectal cancer (Fig. 3).

Discussion

This study is the first to have examined national ethnic differentials in cancer incidence between South Asians and non-South Asians in England as well as the extent to which such differentials were driven by differentials in socioeconomic deprivation. The findings revealed first that both ethnicity and socioeconomic deprivation exerted separate effects on cancer incidence, with the nature of the deprivation gradient being to a large extent ethnic-specific, highlighting the need for ethnic disparities not to be taken as a proxy for socioeconomic inequalities, and *vice versa*, in official statistics and epidemiological studies. Second, despite an increase in total cancer incidence in both South Asians and non-South Asians over two decades, rates among UK South Asians remained low regardless of age. Finally, the study also shows in general weaker or no deprivation gradient for cancer incidence in South Asians.

These substantial findings should be seen in the light of incidence patterns observed in South Asia. Changes in incidence in England-based South Asians may simply mirror secular trends in their countries of origin perhaps, because they share common drivers of, for instance, patterns of

behavior. Although population-based incidence data are patchy for India and Pakistan and absent for Bangladesh, *Cancer Incidence in Five Continents* series,^{19,20} which publishes only data from registries deemed to be reliable, noted that all-cancer incidence was fairly stable in India. In Pakistan, where incidence is estimated on a single cancer registry (South Karachi), cancer incidence was increasing.^{19,20} Lung cancer incidence, mainly caused by tobacco smoking, was much higher among England-based South Asians than in India and slightly higher than in Pakistan,²⁰ but incidence of oral cancer was much lower among South Asians in England than in South Asia over the entire study period (data not presented) consistent with a shift from chewing tobacco to cigarette use among South Asians after settling in the United Kingdom.^{21,22}

Incidence of the main hormone-related cancer, breast, increased among England-based South Asians, resulting in much higher incidence than in the South Asian subcontinent. Similar transitional trends in breast cancer incidence have been reported among Japanese migrants to the United States¹ and migrants to the Netherlands.²³ This recent increase in breast cancer is likely to persist, because fertility rates in South Asians were half those observed in South Asian countries.^{24,25} Similar patterns were also observed for other hormone-related cancers such as uterus, ovary and testis (data not shown). Another striking illustration of the

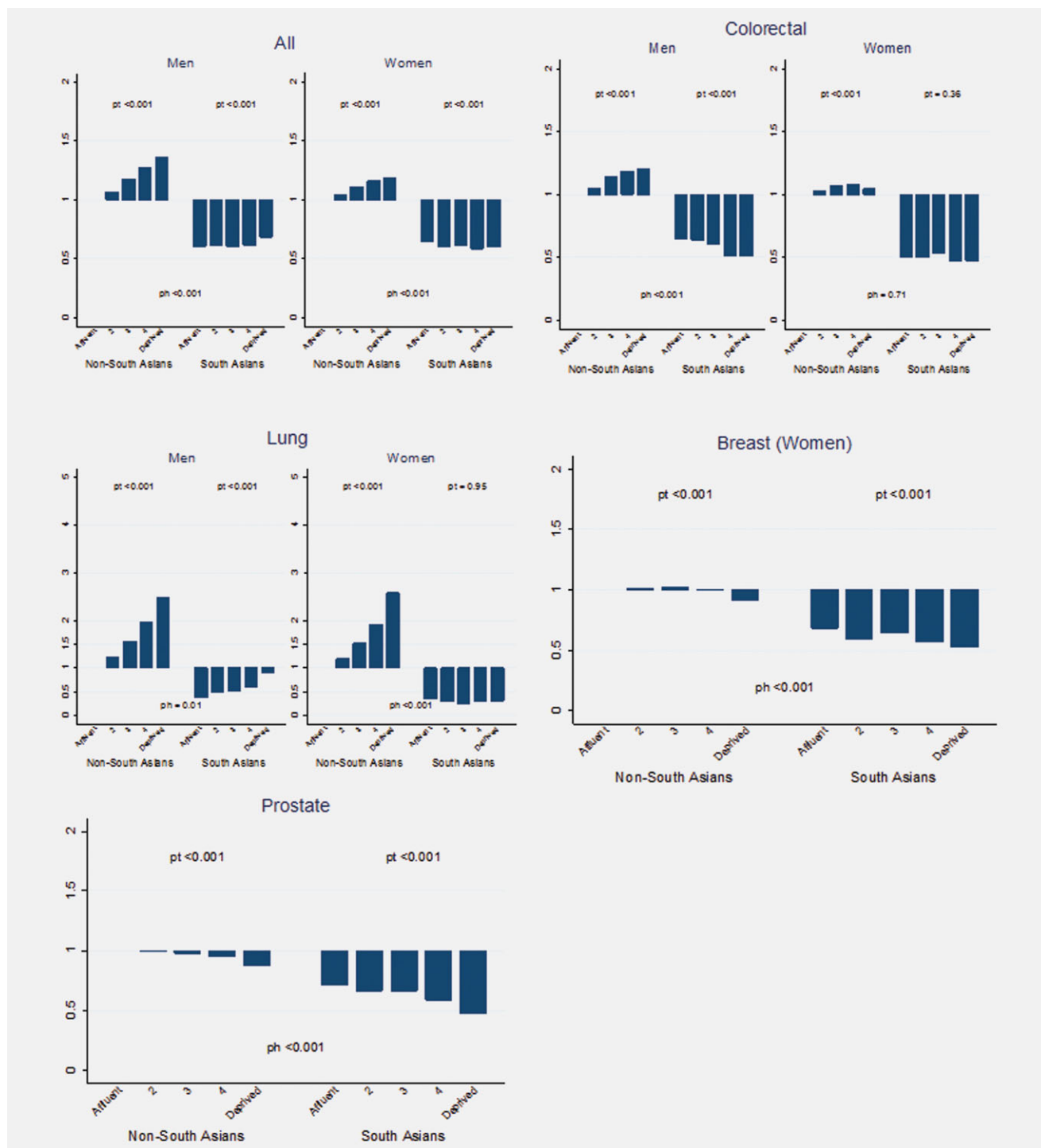


Figure 4. Modeled cancer incidence rate ratios by cancer, sex, ethnic group and deprivation quintile comparing incidence to the most affluent non-South Asian group, in 50-year olds in 2000. p_t corresponds to the p test-for-trend in the incidence rate ratios, by sex, ethnicity and cancer. p_h corresponds to the test for heterogeneity in incidence trends by deprivation between ethnic groups. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

epidemiological transition is the rapid increase in colorectal cancer incidence in England-based South Asians over time, particularly among younger South Asians, possibly reflecting dietary change or greater health-seeking behavior.

The study specifically examined the impact of further adjusting for deprivation. The role of social disadvantage as well as cultural, lifestyle and genetic factors in population differences in cancer patterns is of interest. South Asians

had stronger inverse deprivation gaps than non-South Asians for breast and prostate cancers. However, South Asians had low or absent socioeconomic gradients for all cancers combined, lung and colorectal cancers, in contrast to the general population. The low-cancer incidence rates across all deprivation groups in South Asians coupled with a greater proportion of the South Asian population in the lowest deprivation category from non-South Asians make the ethnic differentials appear to be larger than they actually are. The deprivation gradients here confirm suggestions of such trends in regional data in England as well as patterns in the United States.^{6,7} The study of the incidence of five major cancer sites by ethnic and socioeconomic groups in California also showed inverse deprivation gradients in breast and prostate cancer incidence for all ethnic groups, including Asians and Pacific Islanders for whom the gap was stronger.²⁶ Weaker or no deprivation gap in incidence for colorectal and lung cancers was also observed in US Asians compared to non-Hispanic Whites.⁵ Overall, we show that England-based South Asians have less-pronounced deprivation gaps in cancer incidence than non-South Asians. By contrast, in the United States, a decrease in mortality rates between 1993 and 2001 was seen mostly in people with at least 16 years of education³⁰ with educational gradients in cancer mortality that were greater among Blacks than Whites. It reflects that alongside socioeconomic deprivation and age,²⁹ the effects of ethnicity on particular outcomes are different in different social settings. It is well recognized that ethnicity is a vector of inequalities in awareness, diagnosis, timeliness of and access to treatment.

The possible social factors mediating the transition to high-income pattern of diseases are likely to be cancer-specific. The high prevalence of smoking particularly in the more recent Bangladeshi migrants²¹ may be behind the increasing lung cancer incidence trends in South Asians. The increasing breast cancer incidence in South Asians is likely to reflect changes in reproductive behavior, diet and body size. For prostate cancer, lower access to diagnostic tests with deprivation in South Asians than in the equivalent groups of non-South Asians may explain the steeper inverse deprivation gap in South Asians than non-South Asians. Lower processed and red meat intake in all South Asian socioeconomic groups and/or lower ascertainment in lower South Asian socioeconomic groups may be important when considering the colorectal cancer findings.²⁷ Despite recent progress, breast cancer screening uptake remains much lower in South Asians than in non-South Asians.²⁸ Similarly, the pilot bowel cancer screening programme showed that uptake in South Asians was much lower than in non-South Asians.²⁸

We applied flexible models on all incident cancer cases in South Asians resident in England over a period of 19 years, enabling a more robust estimation of temporal and socioeconomic patterns in incidence than is possible with a solely descriptive approach. The assignment of South Asian ethnicity,

however, differed between the cancer registrations and the general population, but SANGRA has been shown to have high sensitivity (89–96%) and specificity (94–98%) against self-assigned South Asian ethnicity.¹¹ Although ethnicity is now routinely recorded in hospital discharge statistics (Hospital Episode Statistics), it was still missing for 24% of patients diagnosed with a malignant neoplasm during 2002–2006.³¹ Country of birth has been used by others as a proxy, but it cannot identify subsequent generations born in the United Kingdom.

The Townsend score, which is based on four census questions, unemployment, car ownership, household owner occupation and overcrowding,¹⁵ might be less relevant for the South Asian ethnic group. However, whatever the measure used, for example, income or proportion of unemployment, the weight of the most deprived group remains unchanged among the South Asians population and cancer patients (data not presented) with the proportion of most deprived patients by cancer ranging between 35 and 60% in South Asians, while this proportion is rarely over 30% among non-South Asians.

We recognize that social mobility may be differential between South Asians and non-South Asians and that the deprivation quintile of the area of residence at diagnosis may not reflect the life-time exposure. The impact of this on our findings is unclear.

In conclusion, the rapid convergence in the incidence of several cancers noted here between South Asians and non-South Asians, especially among the younger adults, corroborates the recent trends in cancer mortality observed in the UK-born South Asians.¹⁰ In addition, this analysis of incidence shows that the convergence is partially hidden by ethnic differences in deprivation. Such changes have potential implications on the demand for health services. These implications have been integrated in both the 2000 NHS Cancer Plan in England³² and the 2007 strategy to improve access to cancer services.²⁹

In the absence of significant falls in the incidence of cancers associated with tobacco smoking among the South Asians, appropriate preventive measures are required to reduce the long-term risk of such cancers in South Asians.²²

Although socioeconomic inequalities in cancer incidence seem less marked within the South Asian group, over half of South Asians belong to the most deprived category. If cancer incidence continues to rise in South Asians and show similar deprivation gaps to non-South Asians, it is likely to strengthen the inequalities in access to delivery and hence impact of health services such as cancer screening.

Authors' contributions

P.M. had the original idea for the study and applied for the project grant with ISS and MC. P.M., C.M., B.R. and I.D.S.S. designed the study. C.M. conducted the analyses and wrote the first draft of the article with P.M. and B.R. B.R., I.D.S.S.,

P.M. and D.L. helped in the data analysis and interpretation of the results. C.M., P.M., B.R., I.D.S.S., D.L. and M.P.C. provided inputs to the presentation of the results. P.M., B.R., I.D.S.S., D.L. and M.P.C. reviewed final drafts of the work.

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